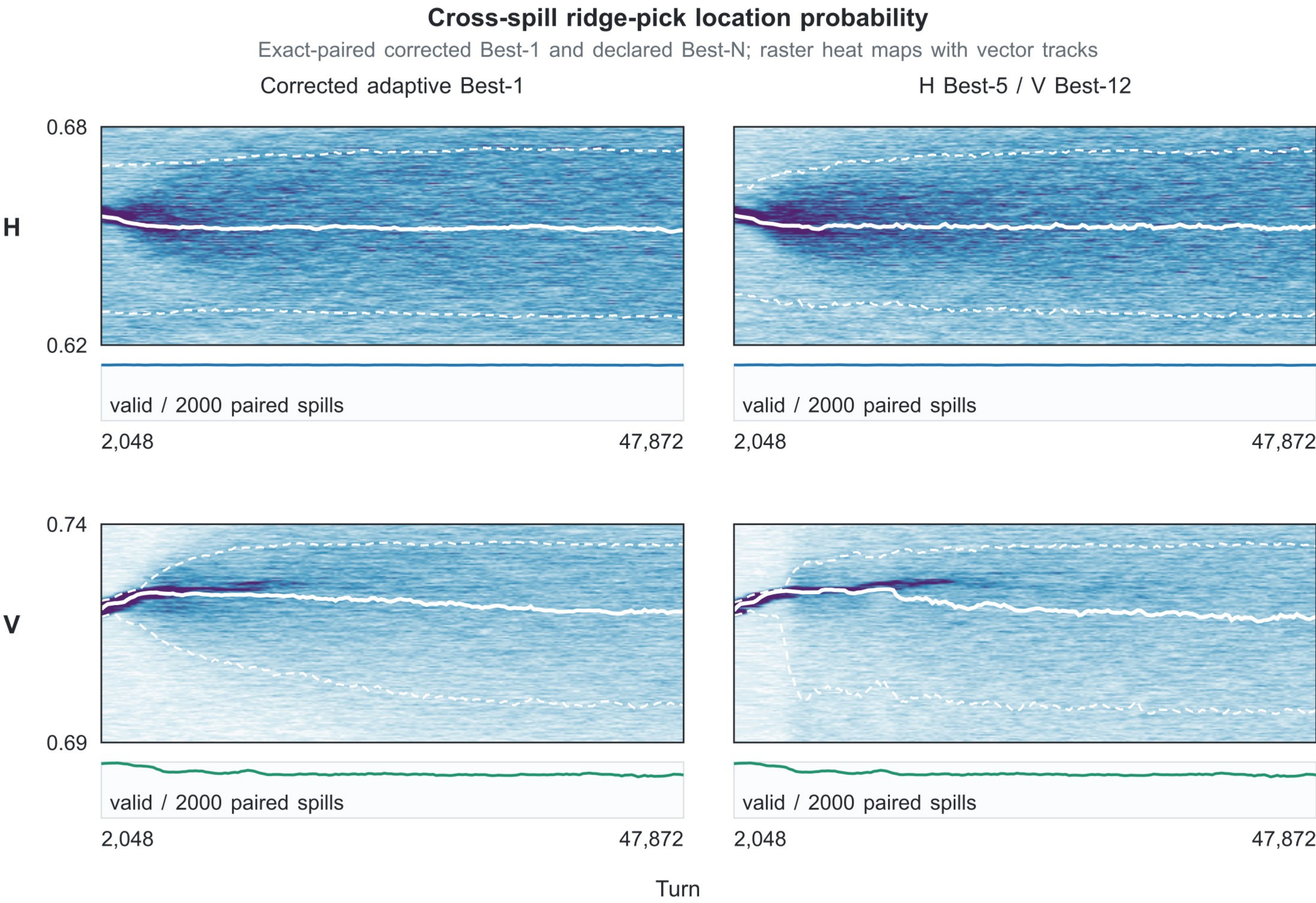


Which BPMs can we trust, spill by spill?

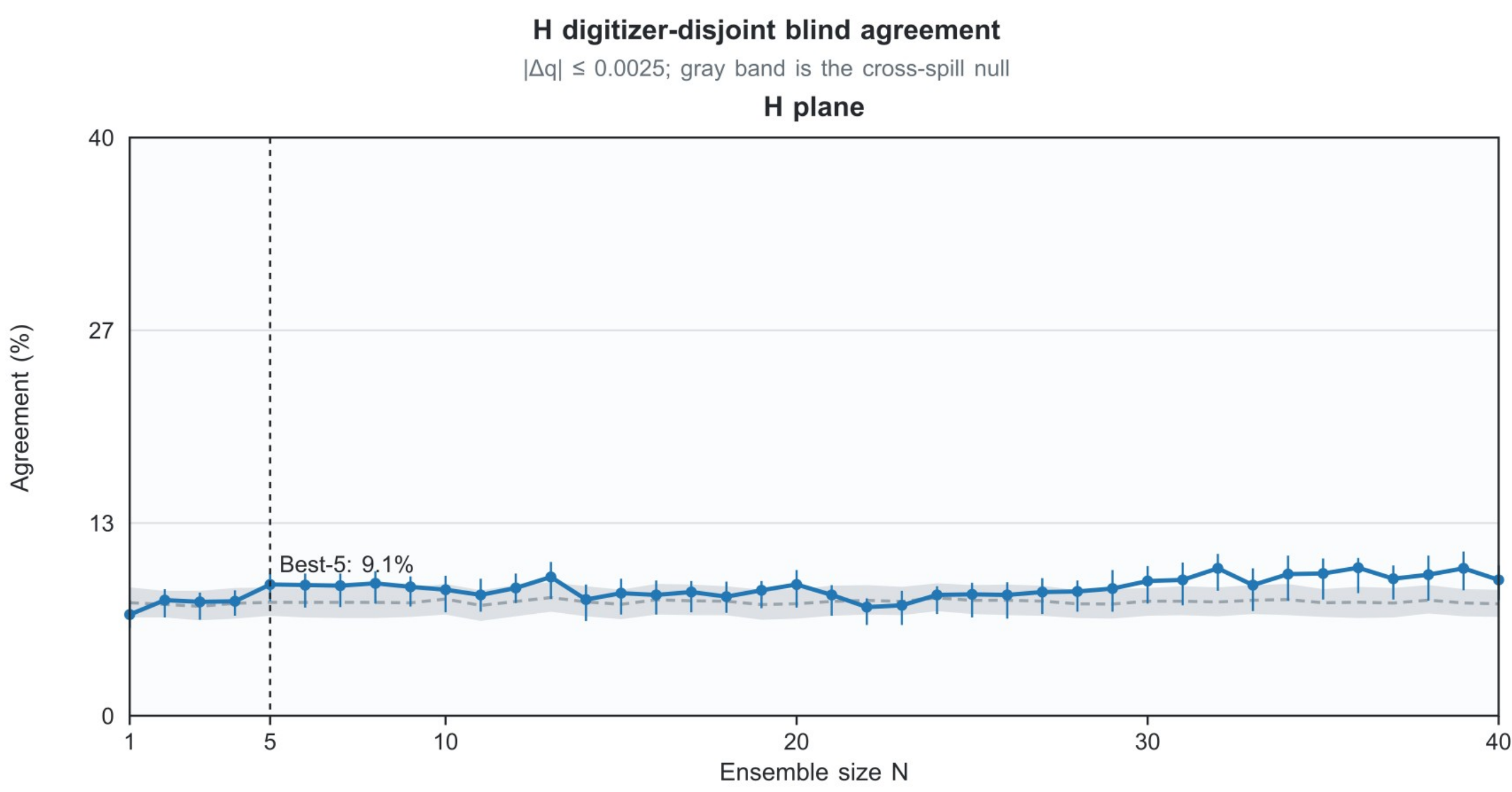
Adaptive turn-by-turn tune analysis in the Mu2e Delivery Ring
Derek Steinkamp | Fermi National Accelerator Laboratory

DOES THE CANDIDATE PERSIST FOR 50,000 TURNS?

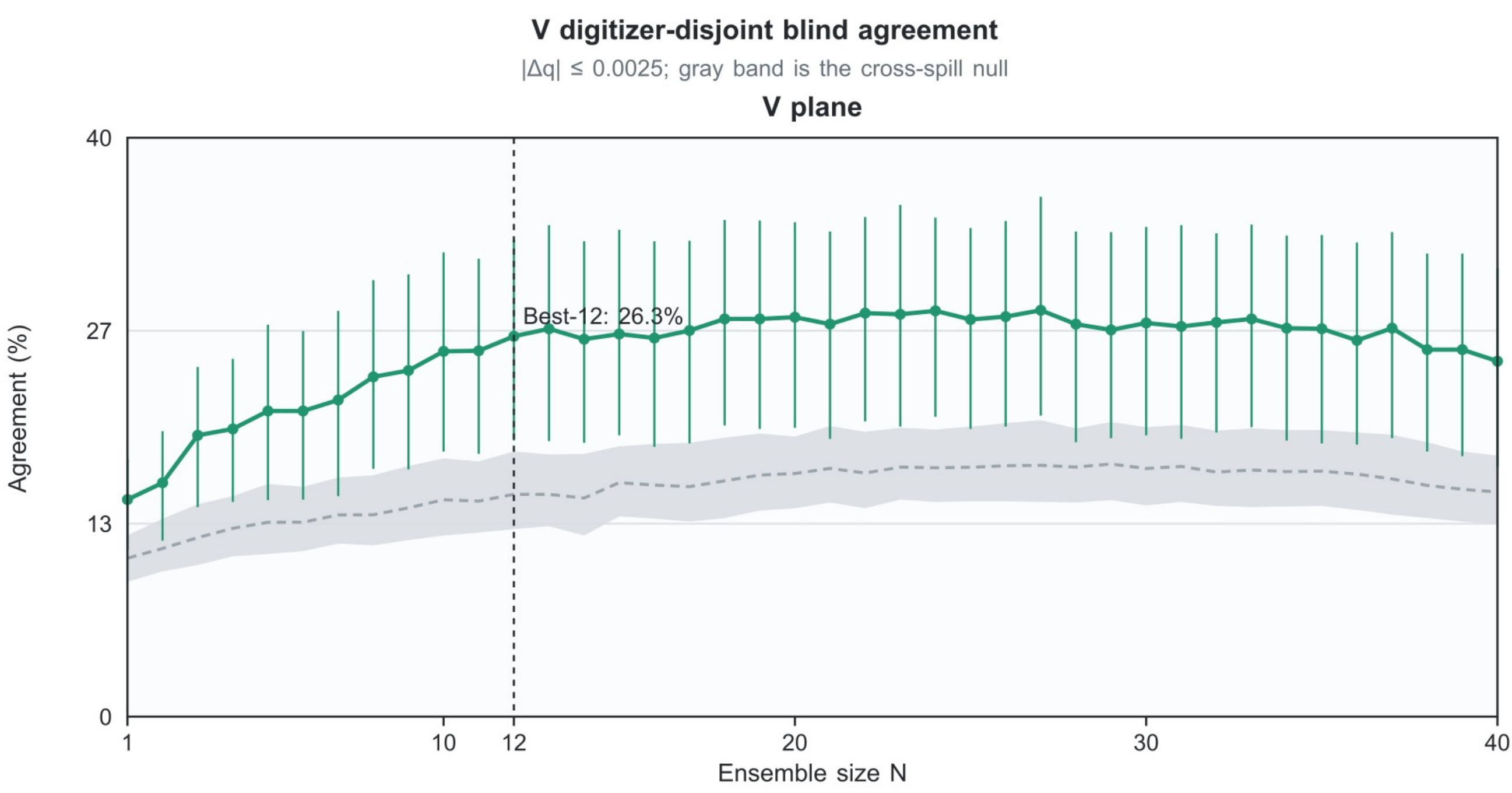


THERE IS NO SINGLE BEST BPM

All 60 H and 60 V sources win at least once. Choose early; test later with held-out digitizers.

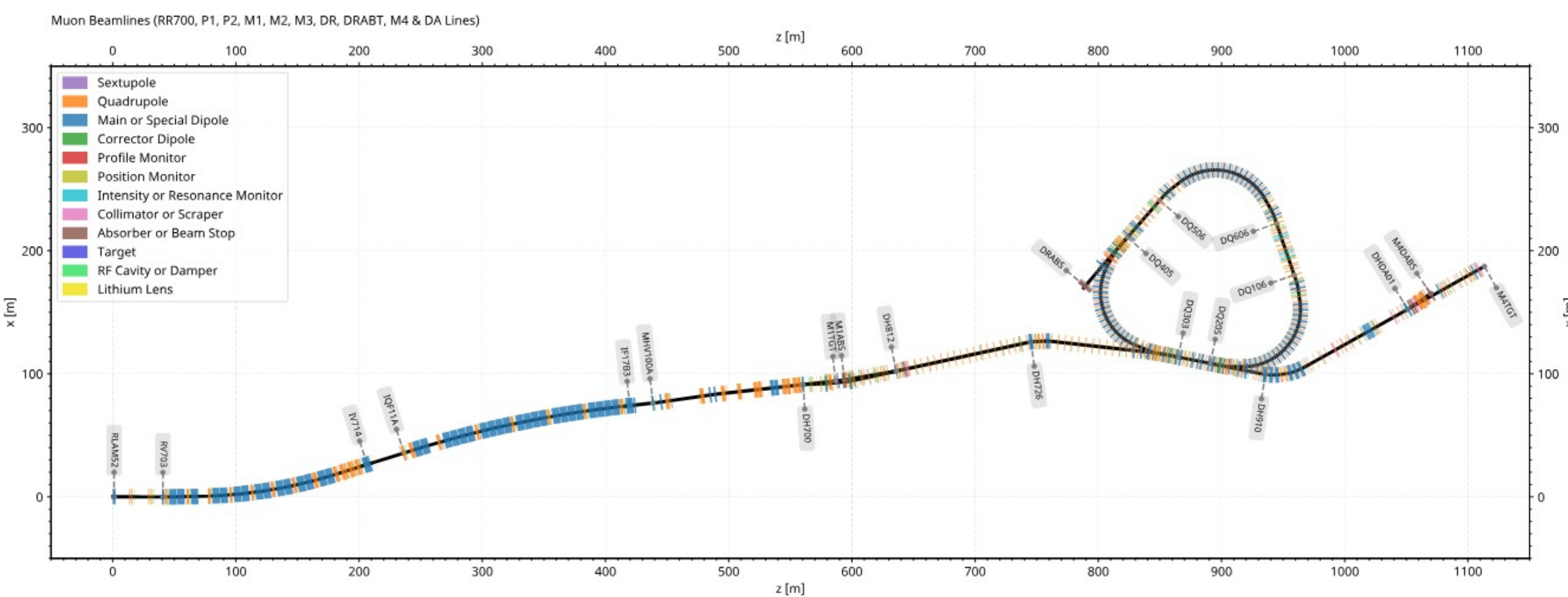


H: Best-5 reached 9.1%. The null band ends at 8.9% - promising, not decisive.



V: Best-12 reached 26.3%. The null band ends at 18.3% - a clear separation.

The loop at right is the Delivery Ring. Its position readouts do not behave equally.



WHAT DID WE LEARN?

H Best-5 narrows the ridge. V Best-12 agrees more strongly. All-channel aggregation remains competitive.

Useful operating points - not universal optima.

Is this the machine tune? Not yet. Next: change the tune on purpose (a controlled quadrupole scan) and ask whether the candidate follows.